Application of Information & Communication Technologies

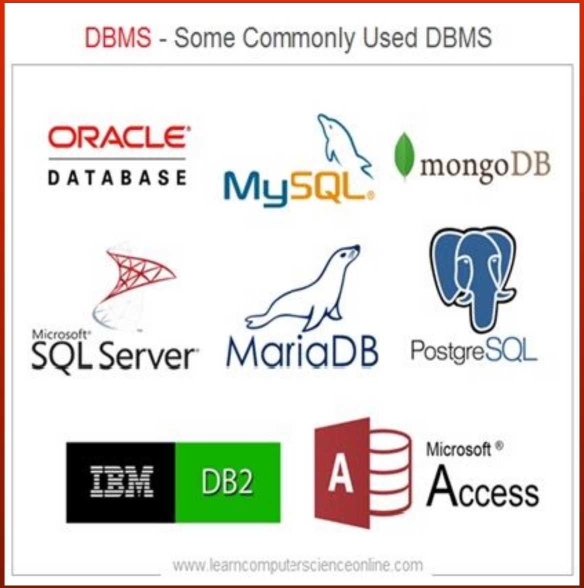
Lecture-18

Recap of Lecture 17

- What is a Database?
- Data, Information, and Tables
- Components of a Database System (DBMS)
- Database vs File System

Overview of Lecture 18

- Types of Databases
- Functions of DBMS
- DBMS Software Examples (MS Access, MySQL)
- Role of Databases in ICT Applications

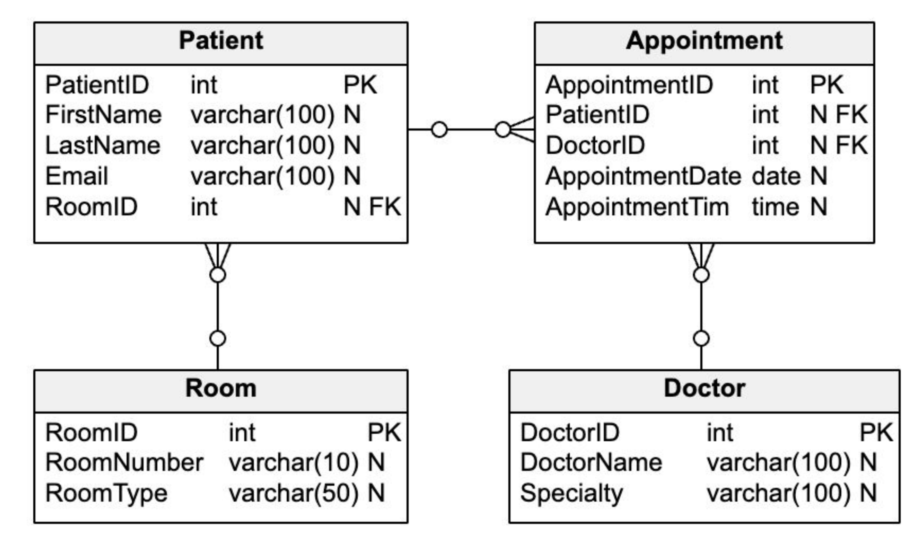


Database management system (DBMS)

- A software system that allows users to create, read, update, and manage data.
- Examples: MS Access, MySQL, Oracle, PostgreSQL.

Types of Databases

Type	Description	Example
Relational	Data in tables with relationships	MySQL, Access
Hierarchical	Parent–child structure	IBM IMS
Network	Complex links between records	IDMS
Object-oriented	Data as objects	db4o
Distributed	Data spread across locations	Google Spanner



Functions of a DBMS

- Data storage, retrieval, and update
- Data security and integrity
- Backup and recovery
- Multi-user access and concurrency control
- Data independence

DBMS Software Examples

Software	Platform	Key Feature
MS Access	Windows	GUI-based for beginners
MySQL	Cross-platform	Free, widely used for web apps
Oracle	Enterprise	High security and scalability
SQLite	Mobile	Lightweight embedded DB

Individuals Involved with a DBMS:

- **Database Designers:** Design the database.
- **Database Developers:** Create the database.
- **Database Programmers:** Write the programs needed to access the database or tie the database to other programs.
- **Database administrators:** Responsible for managing the databases within an organization.
- **Users:** Individuals who enter data, update data, and retrieve information out of the database.

Advantages & Disadvantages of DBMS Approach

Advantages:

- Faster response time.
- Lower storage requirements.
- Easier to secure.
- Increased data accuracy.

Disadvantages:

- Increased vulnerability (backup is essential).

Entity Relationships

One – to – One (1:1)	• Entity relationships (not common) • Each store has a single manager
One – to – Many (O:M)	• Entity relationships (most common) • A supplier supplies more than one product to a company
Many – to – Many (M:M)	• Entity relationships (requires a third table to tie the tables together). • An order can contain multiple products and a product can appear on multiple orders.

Entity Relationships

One – to – One (1:1)

One – to – Many (O:M)

Many – to – Many (M:M)

Summary

- DBMS provides efficient, secure, and scalable data handling.
- Databases support all ICT applications from small systems to cloud platforms.
- Examples: MS Access (local), MySQL (online).

Suggested Reading

- Ch-14, The System Unit: Processing and Memory , “Understanding Computers: Today and Tomorrow, Comprehensive”, 15th Edition by Deborah Morley & Charles S. Parker

- Ch-09, Discovering Computers Fundamentals-Your Interactive Guide -- Gary B Shelly; Misty E Vermaat; Jeffrey J Q